

Section 1.7

Linear Independence

Linear Independence and Dependence

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is linearly independent if the vector equation

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_p \mathbf{v}_p = \mathbf{0}$$

has only the trivial solution.

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if the vector equation

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_p \mathbf{v}_p = \mathbf{0}$$

has at least one nontrivial solution.

Example 1

Determine if the vectors are linearly independent.

$$\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -8 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

Special Cases

One Vector

A one-vector set is automatically linearly independent unless it is the zero vector.

Two Vectors

A set of two nonzero vectors is linearly dependent if and only if one of the vectors is a multiple of the other.

Special Cases

Too Many Vectors

If a set contains more vectors than there are entries in each vector, then the set is automatically linearly dependent.

Zero Vector

A set containing the zero vector is automatically linearly dependent.

Linearly Dependent Sets

An indexed set of two or more vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly dependent if and only if at least one vector in the set is a linear combination of the preceding vectors.

Example 2

For what values of h is $\mathbf{v}_3$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -3 \\ -5 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -3 \\ 9 \\ 15 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ -5 \\ h \end{bmatrix}$$

For what values of h is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly dependent?

Example 3

True or False:

If $\mathbf{x}$ and $\mathbf{y}$ are linearly independent and if $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is linearly dependent, then $\mathbf{z}$ is in $\text{Span}\{\mathbf{x}, \mathbf{y}\}$.

Example 4

Each of the following statements is either true (in all cases) or false (for at least one example). If false, construct a specific example to show the statement is not always true.

If $\mathbf{v}_1, \dots, \mathbf{v}_4$ are in $\mathbb{R}^4$ and $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_4\}$ is linearly dependent.

Example 4 (continued)

Each of the following statements is either true (in all cases) or false (for at least one example). If false, construct a specific example to show the statement is not always true.

If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in $\mathbb{R}^4$ and $\mathbf{v}_2$ is not a scalar multiple of $\mathbf{v}_1$, then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.

Example 4 (continued)

Each of the following statements is either true (in all cases) or false (for at least one example). If false, construct a specific example to show the statement is not always true.

If $\mathbf{v}_1, \dots, \mathbf{v}_5$ are in $\mathbb{R}^5$ and $\mathbf{v}_3 = \mathbf{0}$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_5\}$ is linearly dependent.

Example 4 (continued)

Each of the following statements is either true (in all cases) or false (for at least one example). If false, construct a specific example to show the statement is not always true.

If $\mathbf{v}_1, \dots, \mathbf{v}_4$ are in $\mathbb{R}^4$ and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is also linearly dependent.

Example 5

Use as many columns of A as possible to construct the matrix B with the property that $B\mathbf{x} = \mathbf{0}$ has only the trivial solution.

$$\begin{bmatrix} 3 & -4 & 10 & 7 & -4 \\ -5 & -3 & -7 & -11 & 15 \\ 4 & 3 & 5 & 2 & 1 \\ 8 & -7 & 23 & 4 & 15 \end{bmatrix}$$
